Smart Retail Customer Analysis & Prediction: An RFM-Based Segmentation and Purchase-Likelihood Framework

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Abstract—Retail businesses generate large volumes of raw, unorganised transaction data that is rarely usable in its original form. This paper presents Smart Retail Customer Analysis & Prediction, an end-to-end framework that cleans raw transaction records, derives Recency, Frequency and Monetary (RFM) features extended with product-category diversity and purchase-frequency trend, segments customers with K-Means clustering, and predicts future purchase likelihood and customer value using Logistic Regression, Random Forest and XGBoost. Segmentation and prediction are linked in one pipeline, and both are delivered through a Streamlit/Power BI dashboard that retailers without a data-science background can use. The system is implemented in Python with Pandas, NumPy, Matplotlib/Seaborn and Scikit-learn, with MySQL/SQLite storage. We describe the data-preparation rules, feature definitions, a time-aware evaluation protocol that prevents label leakage, and the dashboard design. The pipeline is executed end to end on a synthetic transaction dataset to verify the implementation and to produce example outputs; evaluation on real retail data is left for future work.

Keywords—retail analytics, customer segmentation, RFM, K-Means, purchase prediction, XGBoost, dashboard.

I. INTRODUCTION

Modern retailers record every purchase, yet most of this data remains in raw tables that are reviewed through simple totals and averages. Such summaries hide how customers differ: two customers with equal annual spending may buy very differently, one making many small purchases and another a single large one. Without a structured view of customer behaviour, retailers tend to send the same promotion to everyone, which lowers the relevance of marketing and wastes budget.

Two analytical tasks address this problem. Segmentation groups customers by similar behaviour so that each group can be treated differently. Prediction estimates which customers are likely to buy again and how much they may spend. In practice the two are often handled separately, and the results are rarely presented in a form that store managers can act on. This paper proposes Smart Retail Customer Analysis & Prediction, a unified workflow that converts raw transactions into analysis-ready customer features, segments customers, predicts future purchasing, and presents the results on an interactive dashboard.

Retailers collect large amounts of customer and transaction data, but manual analysis becomes impractical across thousands of customers because purchase frequency, spending and product preferences interact in complex ways. Without segmentation, marketing is untargeted, and without prediction, retailers cannot identify which customers are likely to buy again. The objectives of this work are: (1) analyse customer behaviour from purchase history, frequency, spending and product preferences; (2) segment customers into meaningful groups such as high-value, frequent, at-risk and price-sensitive buyers; (3) predict the likelihood and value of a customer's future purchases; (4) deliver the results through a simple dashboard; and (5) support data-driven decisions in personalised marketing and sales planning.

The main contributions are: (1) a documented cleaning and validation procedure for retail transaction data; (2) an extended feature set that adds category diversity and frequency trend to standard RFM; (3) a pipeline that links K-Means segments to supervised purchase-likelihood and customer-value models; (4) a time-aware evaluation protocol that avoids leakage between features and labels; and (5) a retailer-oriented dashboard that produces action lists such as at-risk alerts and targeted promotion lists.

The remainder of the paper is organised as follows. Section II reviews related work, Section III describes the data resources and preparation, Section IV presents the proposed framework, Section V discusses the demonstration run, evaluation protocol and results, and Section VI concludes. The output figures of the demonstration run are presented after the conclusion, followed by the references.

II. LITERATURE SURVEY

RFM analysis summarises purchasing behaviour through Recency, Frequency and Monetary value and has long been used for customer ranking and valuation [1]. Chen et al. applied RFM with data-mining techniques to a UK online-retail dataset and showed that RFM-based clustering yields interpretable customer groups [2]. Because RFM uses only three variables, later work has extended it. Mahfuza et al. proposed LRFMV, which adds customer relationship Length and purchase Volume, and compared K-Means, K-Medoids and Mini-Batch K-Means on superstore data, reporting clearer profit-to-quantity relationships and the most precise clusters with K-Means [3]. Ullah et al. introduced RFMT, adding a Time dimension, and used

cluster validation with majority voting across several algorithms on e-commerce data, reporting improved cluster validity over standard RFM [4]. A recent survey by Chavhan et al. compared K-Means, hierarchical and fuzzy C-means approaches on online-retail datasets and noted that fuzzy C-means achieved better silhouette scores at higher computational cost [5]. Table I summarises these studies.

TABLE I. SUMMARY OF RELATED WORK

Work	Method	Limitation
Mahfuza *et al.* [3]	LRFMV with K-Means, K-Medoids, Mini-Batch K-Means	Volume feature is store-specific; no purchase prediction
Ullah *et al.* [4]	RFMT with cluster validation and majority voting	Segmentation only; no forward-looking prediction
Chavhan *et al.* [5]	Survey of RFM clustering (K-Means, hierarchical, fuzzy C-means)	Survey level; no end-to-end system

For prediction, Logistic Regression provides a transparent baseline for binary outcomes [6], Random Forest averages many decorrelated trees to model non-linear interactions [7], and XGBoost provides regularised gradient-boosted trees that often perform strongly on tabular data [8]. **Research gap.** The surveyed segmentation studies use a narrow feature set, stop at descriptive clustering, and rarely connect segments to a forward-looking model or to a tool suitable for small and medium retailers. This work addresses these gaps by combining extended features, segmentation and prediction in a single workflow with an accessible dashboard.

III. DATA RESOURCES

A. Data Acquisition and Schema

The input is a table of retail transactions with invoice identifier, customer identifier, purchase date, product category or description, quantity and unit price. Data may come from a CSV export or a retail database. A publicly available transactional dataset, such as the UCI Online Retail dataset [9], can be used for development and benchmarking because it has this schema.

B. Data Cleaning and Validation

Cleaning converts raw records into a consistent dataset, following standard data-preparation practice [10]. Data types are corrected and dates are parsed. Rows with missing customer identifiers are removed because they cannot be attributed to a customer. Duplicate rows are examined before deletion so that genuine repeat purchases are kept. Cancelled invoices and negative quantities (returns) are separated rather than silently dropped, since they carry information about customer behaviour. Impossible values, such as non-positive prices, are removed, and extreme quantities or prices are flagged using the interquartile-range rule and winsorised only when they are clearly erroneous, because large purchases by wholesale customers can be genuine. Text fields such as category names are normalised. The number of rows before and after each step, with the reason for each removal, is logged so that data-quality improvements can be separated from changes that might bias the analysis.

C. Demonstration Dataset

To verify the pipeline end to end, it was executed on a synthetic transaction dataset generated with a fixed random seed. The generator simulates 3,000 customers of four latent behaviour types (high-spending loyal, frequent small-basket, occasional, and lapsing) buying from eight product categories between 1 January 2023 and 31 December 2024, with seasonality and with injected data-quality problems (missing customer identifiers, duplicate rows, returns and zero prices). The raw table has 76,782 rows. Training features use the 12 months before a cutoff of 1 April 2024 with labels from the following 90 days; testing uses a later cutoff of 1 July 2024 with its own 90-day labels. Because the data are synthetic, the numbers reported in Section V demonstrate that the pipeline works and are not evidence of performance on real retail data. The reference real-world dataset remains the UCI Online Retail data [9].

IV. PROPOSED FRAMEWORK: SMART RETAIL CUSTOMER ANALYSIS & PREDICTION

The workflow in Fig. 1 has seven stages: data acquisition, cleaning, feature engineering, scaling, segmentation and prediction (which share the same features), evaluation, and dashboard delivery. Each stage is implemented as a separate module so that it can be tested independently.

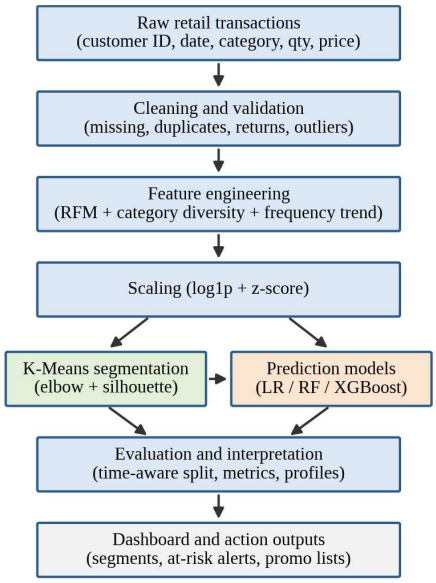


Fig. 1. Proposed end-to-end workflow.

A. Feature Engineering

Features are computed per customer over an observation window ending at a reference date T. The reference date and window length are recorded for reproducibility. Let t_{last} be a customer's latest purchase date in the window, N the number of distinct invoices, and q_i and p_i the quantity and unit price of line i. Then

$$R = T - t_{\text{last}} \text{ (days)} \quad (1)$$

$$F = N \quad (2)$$

$$M = \Sigma_i \, q_i \cdot p_i \quad (3)$$

Two extended features capture behaviour that RFM misses. Category diversity is the Shannon entropy of the customer's spending shares s_c across product categories,

$$D = -\Sigma_c \, s_c \ln s_c \quad (4)$$

and the frequency trend β is the least-squares slope of the customer's monthly purchase count against month index, so that $\beta > 0$ indicates increasing and $\beta < 0$ declining engagement. Average basket value M/F is also derived. Table II lists the resulting features.

TABLE II. CUSTOMER FEATURES

Feature	Definition	Purpose
Recency (R)	Days since last purchase	Detect lapsing customers
Frequency (F)	Distinct invoices in window	Measure loyalty
Monetary (M)	Total spend in window	Measure value
Basket value	M / F	Separate bulk from small buyers
Diversity (D)	Entropy of category shares	Product-preference breadth
Trend (β)	Slope of monthly purchases	Early signal of churn or growth

Because RFM variables are right-skewed and K-Means is distance based, every feature is transformed with $\log(1+x)$ where it is non-negative and then standardised to zero mean and unit variance.

B. Customer Segmentation

K-Means partitions the scaled feature vectors into K clusters by minimising the within-cluster sum of squares [11],

$$J = \Sigma_{i=1}^{n} \, \|\mathbf{x}_i - \boldsymbol{\mu}_{c(i)}\|^2 \quad (5)$$

where $c(i)$ is the cluster assigned to customer i and $\boldsymbol{\mu}_k$ is the centroid of cluster k. The number of clusters is chosen by comparing the elbow curve of J with the mean silhouette score [12],

$$s(i) = (b(i) - a(i)) \, / \, \max\{a(i), b(i)\} \quad (6)$$

where $a(i)$ is the mean distance of point i to its own cluster and $b(i)$ the smallest mean distance to another cluster, together with business interpretability. Each cluster is then profiled by size and by mean recency, frequency, monetary value, diversity and trend. Descriptive names such as high-value, frequent, at-risk or price-sensitive are assigned by reviewing these profiles; they are interpretations and not outputs of the algorithm. Stability is checked by repeating K-Means with different random seeds, and clusters too small to support decisions are merged or reviewed.

C. Purchase Prediction

Prediction is a supervised task kept separate from segmentation. Two targets are defined. The classification target y indicates whether the customer makes at least one purchase in a 90-day horizon after a cutoff date T; the value target is the customer's spend in the same horizon. Features are computed only from data before T, and labels only from the horizon after T (Fig. 2), so that no future information leaks into the features. Evaluation uses a later cutoff than training, and any overlap of customers between training and test periods is reported.

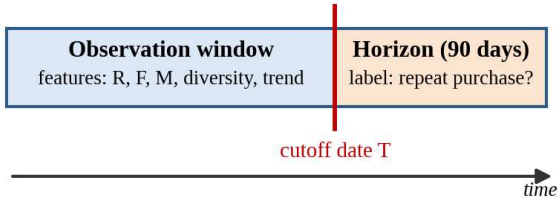


Fig. 2. Time-aware construction of features and labels.

Logistic Regression serves as the interpretable baseline. Random Forest and XGBoost model non-linear interactions among the features, and the K-Means segment label is added as an input feature to test whether segmentation improves prediction. For customer value, regression variants of Random Forest and XGBoost are used. Class imbalance is assessed and handled by class weights or threshold tuning rather than by resampling across the time boundary. Hyperparameters are tuned with time-ordered cross-validation inside the training period only. Table III lists the planned search space. Searching is performed only on the training period with time-ordered folds, and the held-out period is used once for the final comparison.

TABLE III. PLANNED HYPERPARAMETER SEARCH SPACE

Model	Parameters searched
Logistic Regression	Regularisation strength C; L1/L2 penalty; class weight
Random Forest	Number of trees; maximum depth; minimum samples per leaf
XGBoost	Learning rate; number of rounds; maximum depth; subsample ratio; L2 regularisation
K-Means	K = 2 to 8; 10+ random initialisations per K

D. Algorithm Summary

Algorithm 1 summarises the complete pipeline. Steps 1–3 prepare the data, steps 4–5 perform segmentation, steps 6–8 perform prediction, and step 9 publishes outputs.

ALGORITHM 1. SMART RETAIL CUSTOMER ANALYSIS AND PREDICTION

#	Step
1	Load transactions; parse dates and fix data types; log row counts.
2	Remove invalid rows; separate returns; handle duplicates and outliers by documented rules.
3	Choose cutoff T and window; compute R, F, M, basket value, D and β per customer.
4	Apply $\log(1+x)$ and standardisation to the feature matrix.
5	Run K-Means for a range of K; select K by elbow, silhouette and interpretability; profile and name segments.
6	Build labels from the 90-day horizon after T; add the segment label as a feature.
7	Train Logistic Regression, Random Forest and XGBoost with time-ordered cross-validation.
8	Evaluate on a later period against baselines; choose the final model and decision threshold.
9	Store results in MySQL/SQLite; refresh the dashboard and action

#	Step
	lists.

E. System Architecture and Implementation

The system uses a layered architecture (Fig. 3). Pandas and NumPy handle data structuring and feature computation, Matplotlib and Seaborn produce visualisations [13], [14], and Scikit-learn provides K-Means, Logistic Regression and Random Forest, with XGBoost for gradient boosting [15]–[17]. MySQL or SQLite stores cleaned transactions and derived customer summaries so that the analytical stages can be rerun without repeating ingestion. A Streamlit or Power BI dashboard reads the stored summaries and model outputs. The code is organised into ingestion, cleaning, feature, segmentation, prediction and dashboard modules, each with its own configuration (reference date, window, horizon, random seed), so that a different retailer's data can be analysed by changing configuration rather than code.

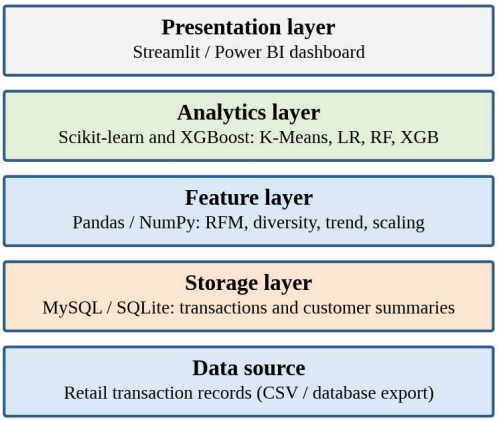


Fig. 3. Layered system architecture.

F. Dashboard and Action Outputs

The dashboard (Fig. 4) shows key indicators, segment sizes and profiles, purchase trends, and customer-level predictions. It also generates action-oriented outputs: a targeted promotion list for each segment, alerts for at-risk customers whose predicted purchase probability is low or whose frequency trend is declining, and flags for high-value customers. Access to customer identifiers is restricted and exported reports omit unnecessary personal information.

Four dashboard views are planned. The *overview* page shows customer count, repeat-purchase rate, revenue and the share of at-risk customers. The *segments* page shows the size of each segment and its mean recency, frequency, monetary value, diversity and trend, with the segment names chosen by the analyst. The *trends* page plots monthly purchases and revenue by segment and category. The *predictions* page lists customers with their purchase probability and expected value, with filters by segment, so that a store manager can export a call list or promotion list in a few clicks. All views are designed for users without a data-science background, with plain-language labels and short explanations beside each chart.

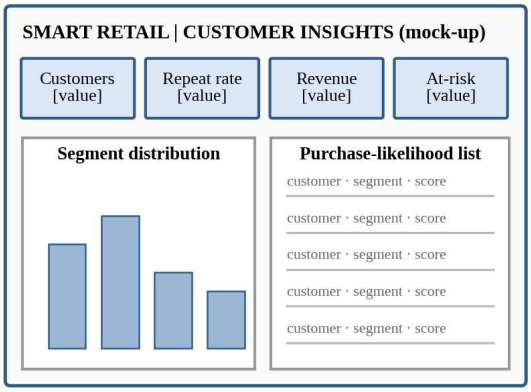


Fig. 4. Dashboard layout mock-up; values are placeholders, not results.

V. RESULT AND DISCUSSION

A. Evaluation Protocol

Segmentation and prediction answer different questions and are evaluated separately. Clustering has no ground-truth label, so internal measures (within-cluster sum of squares and mean silhouette score) are combined with a review of segment profiles and a seed-stability check. For classification, models are compared with two baselines: a majority-class predictor and a simple recency rule. Precision, recall, F1-score and ROC-AUC are reported on the held-out later period,

$$F1 = 2 \cdot P \cdot R_c / (P + R_c) \qquad (7)$$

where P is precision and R_c is recall. Confusion matrices show the cost of false positives (promotions sent to customers who would have bought anyway) and false negatives (missed at-risk customers). For customer value, MAE and RMSE are reported. An ablation compares RFM-only features with the extended feature set, and models with and without the segment label, to test whether the additions help. Dataset size, date coverage, missing-value handling, random seeds and software versions are recorded.

Computational cost. K-Means has cost proportional to the number of customers, clusters, features and iterations, and the feature matrix has only six columns, so segmentation is fast even for hundreds of thousands of customers. Tree-based models scale with the number of customers and trees, and the full pipeline is expected to run on a standard laptop for datasets of the size used by small and medium retailers. This matters for the target users, who typically cannot afford heavy infrastructure.

B. Experimental Setup

The demonstration run uses the synthetic dataset described in Section III-C. XGBoost was not installed in the demonstration environment, so scikit-learn's histogram gradient boosting (marked GB*) stands in for it; the provided script uses XGBoost automatically when it is installed. For reproducibility, the repository for the study contains the cleaning rules, feature code, the pipeline script, a fixed seed, and a script that regenerates every table and figure. Reported values state the dataset, observation window and evaluation period.

C. Data Cleaning and Segmentation

Cleaning reduced the table from 76,782 to 73,748 rows: 942 rows lacked a customer identifier, 453 were exact duplicates, 1,494 were returns separated from sales, and 145 had non-positive price or quantity. The training and test sets contain 2,613 and 2,694 active customers, with 69.0% and 62.4% of them purchasing in the 90-day horizon. For K-Means, the mean silhouette score was highest at $K = 2$ (0.261) and was 0.236 at $K = 4$ (Fig. 5). $K = 4$ was chosen because two groups are too coarse for targeted marketing; this shows why business interpretability is considered alongside the silhouette score. Table IV profiles the four segments, and Fig. 6(a) shows their sizes.

TABLE IV. SEGMENT PROFILES (MEAN VALUES, SYNTHETIC DATA)

Segment	Size	*R*	*F*	*M*	*D*	Trend
High-value	559	51.6	5.2	1,683	1.08	0.027
Cooling	554	78.3	7.5	756	1.25	−0.056
Active regular	753	21.2	5.9	490	1.18	0.071
At-risk	747	133.6	1.6	165	0.59	0.007

The high-value segment has the highest spend; the at-risk segment has the longest recency, lowest frequency and the narrowest category range; the cooling segment still buys often but has a high recency and a declining trend ($\beta < 0$); and the active regular segment is recent with moderate spend. These names were assigned by reviewing the profiles.

D. Prediction Results

TABLE V. 90-DAY PURCHASE PREDICTION ON THE HELD-OUT PERIOD

Model	Precision	Recall	F1	ROC-AUC
Majority baseline	0.624	1.000	0.768	0.500
Recency rule ($R \leq 60$)	0.731	0.707	0.719	0.671
Logistic Regression	0.776	0.765	0.771	0.758
Random Forest	0.766	0.801	0.783	0.766
Gradient Boosting*	0.706	0.882	0.784	0.756

Because 62.4% of test customers purchase, a model that predicts "will buy" for everyone already reaches F1 = 0.768, so F1 alone is a weak yardstick here and ROC-AUC is more informative. Random Forest achieved the highest AUC (0.766), followed by Logistic Regression (0.758) and gradient boosting (0.756), all clearly above the recency rule (0.671) (Fig. 6(b)). Differences between the three models are small and no significance test was run. The ablation in Table VI shows that extended features raised AUC from 0.752 to 0.763, and adding the segment label gave a further small change to 0.766. Basket value, monetary value and frequency were the most important features (Fig. 7). For customer value, Random Forest regression gave MAE 193.5 and RMSE 295.6, against 242.5 and 342.9 for predicting the training mean.

TABLE VI. ABLATION WITH RANDOM FOREST (SYNTHETIC DATA)

Feature set	F1	ROC-AUC
RFM only	0.775	0.752
Extended features	0.782	0.763
Extended + segment	0.783	0.766

Fig. 8 shows the dashboard populated with the outputs of this run, including the list of at-risk customers with low predicted purchase probability, which is the kind of action list a retailer would use for early outreach.

E. Expected Applications

The framework supports several retail decisions. *Targeted promotions:* each segment receives offers suited to its behaviour, for example loyalty rewards for high-value customers and discount codes for price-sensitive ones, instead of one promotion for all. *Retention:* customers with a declining frequency trend or a low predicted purchase probability are listed for early outreach before they lapse. *Category planning:* comparing category diversity and preferences across segments shows which product groups drive each segment and where cross-selling is possible. *Sales planning:* predicted customer value aggregated over segments gives a forward-looking estimate that complements historical sales reports. These uses are decision support; final actions should be reviewed by retail staff.

F. Limitations and Ethical Considerations

Historical purchasing patterns may not persist because of seasonality, price changes and one-off purchases, and transaction histories may be incomplete. Segments can reflect data-collection or business-policy effects rather than stable customer traits, and predicted scores guide decisions rather than guarantee behaviour. Targeting rules should be reviewed by retail staff and monitored for unintended exclusion or over-targeting of customer groups. Customer identifiers should be pseudonymised in stored summaries and dashboards. The reported numbers come from synthetic data and must not be read as real-world performance.

VI. CONCLUSION AND FUTURE SCOPE

This paper presented Smart Retail Customer Analysis & Prediction, a framework that turns raw retail transactions into structured customer features, K-Means segments, purchase-likelihood and value predictions, and a dashboard with action-oriented outputs. Its design keeps each step interpretable and evaluates prediction separately from segmentation using a time-aware protocol. A demonstration run on synthetic data verified the pipeline from cleaning to dashboard outputs. Future work includes running the pipeline on a documented real retail dataset and reporting measured results, comparing K-Means with K-Medoids and fuzzy C-means, adding explanation methods such as feature-importance analysis for the predictive models, and testing whether segment-targeted actions improve retention or revenue in practice.

ACKNOWLEDGMENT

The authors thank their project guide, Ms. Prathusha, Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation, for guidance during this work.

OUTPUTS OF THE DEMONSTRATION RUN

Figs. 5–8 are generated directly by the pipeline script from the synthetic demonstration run described in Section V. They show the cluster-selection curves, the segment sizes and ROC curves, the feature importance of the Random Forest model, and the dashboard populated with the run's results.

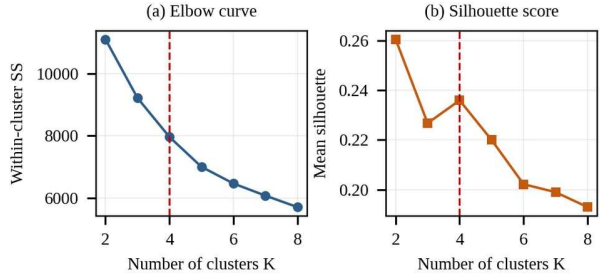


Fig. 5. Elbow curve and mean silhouette score for K = 2 to 8 (synthetic data). The dashed line marks the chosen K = 4.

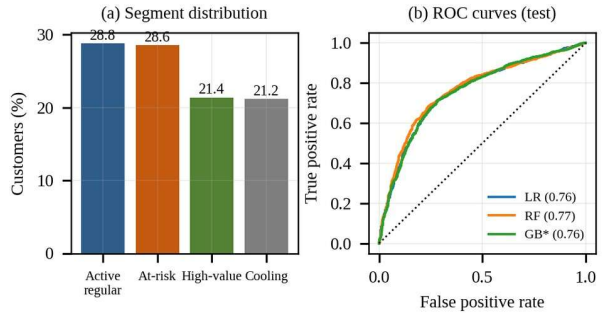


Fig. 6. (a) Segment distribution and (b) ROC curves of the prediction models on the held-out period; AUC in brackets (synthetic data).

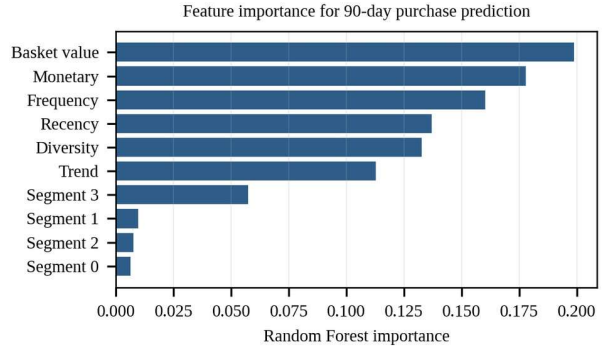


Fig. 7. Random Forest feature importance (synthetic data).

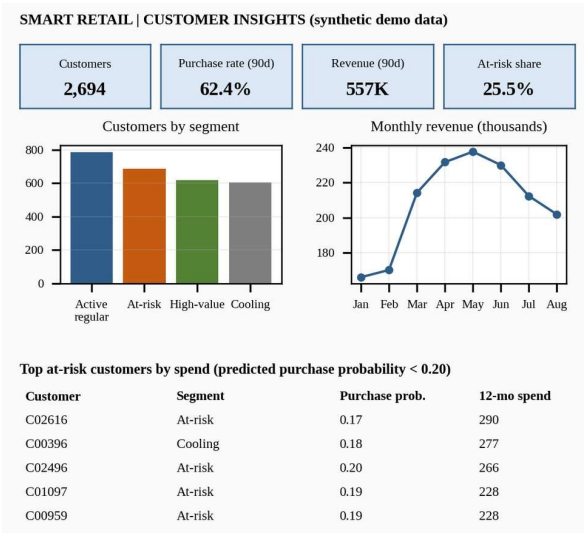


Fig. 8. Dashboard output generated from the demonstration run (synthetic data).

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